

THE HEART

Department of anatomy

First Part

Thorax (4)

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Lecture points

➤ The Middle Mediastinum.

➤ The Pericardium.

- Fibrous pericardium.
- Serous Pericardium.
- Pericardial Cavity.
- Pericardial Sinuses.
- Nerve supply.
- Blood supply.
- Pericardiocentesis.

➤ Location of the heart .

➤ Cardiac Orientation.

➤ Base of the heart.

➤ Anterior surface of the heart.

➤ Diaphragmatic surface.

➤ The Border and margin of the heart.

➤ External Sulci (groove) .

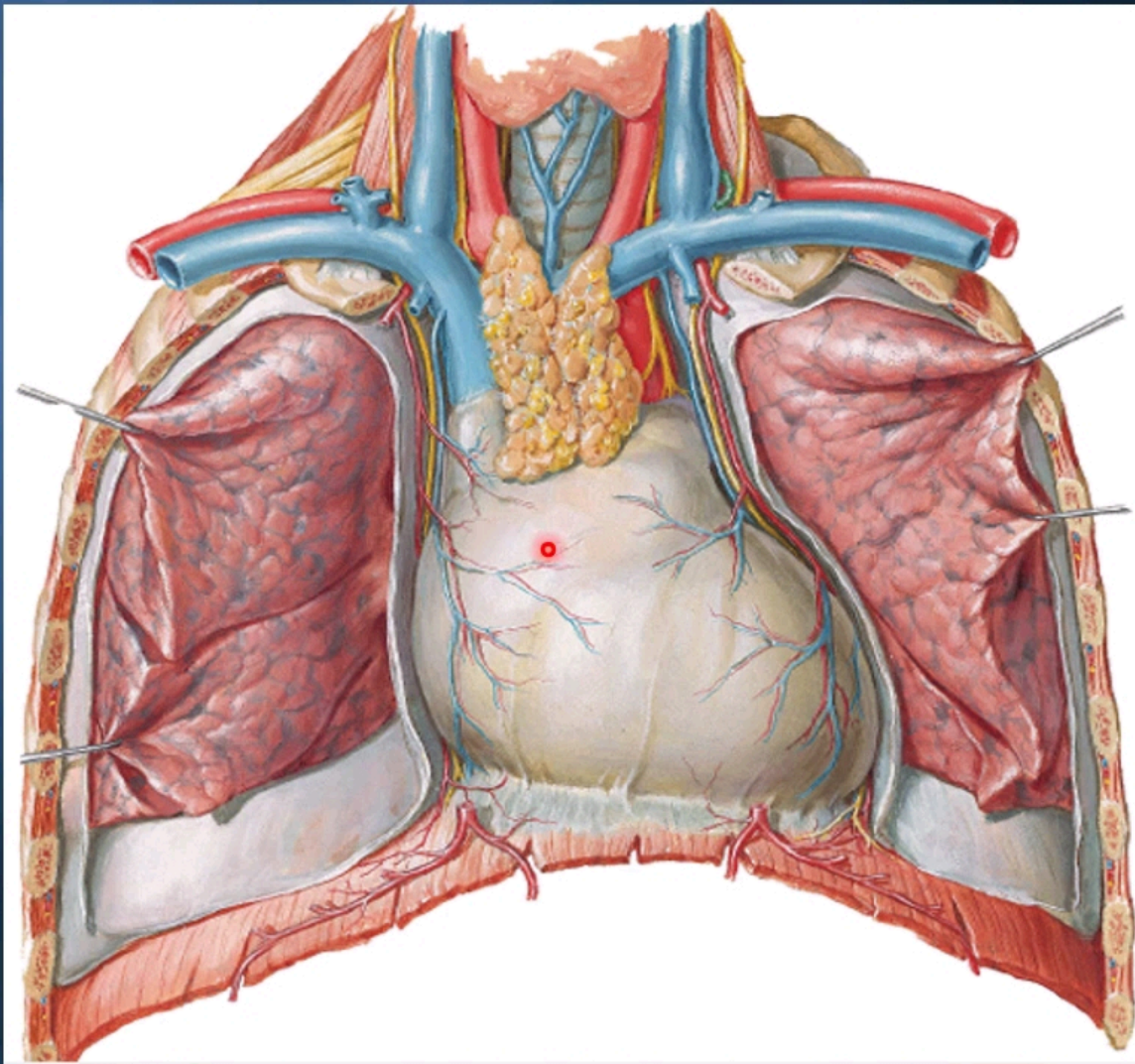
➤ Internal Anatomy of the Heart

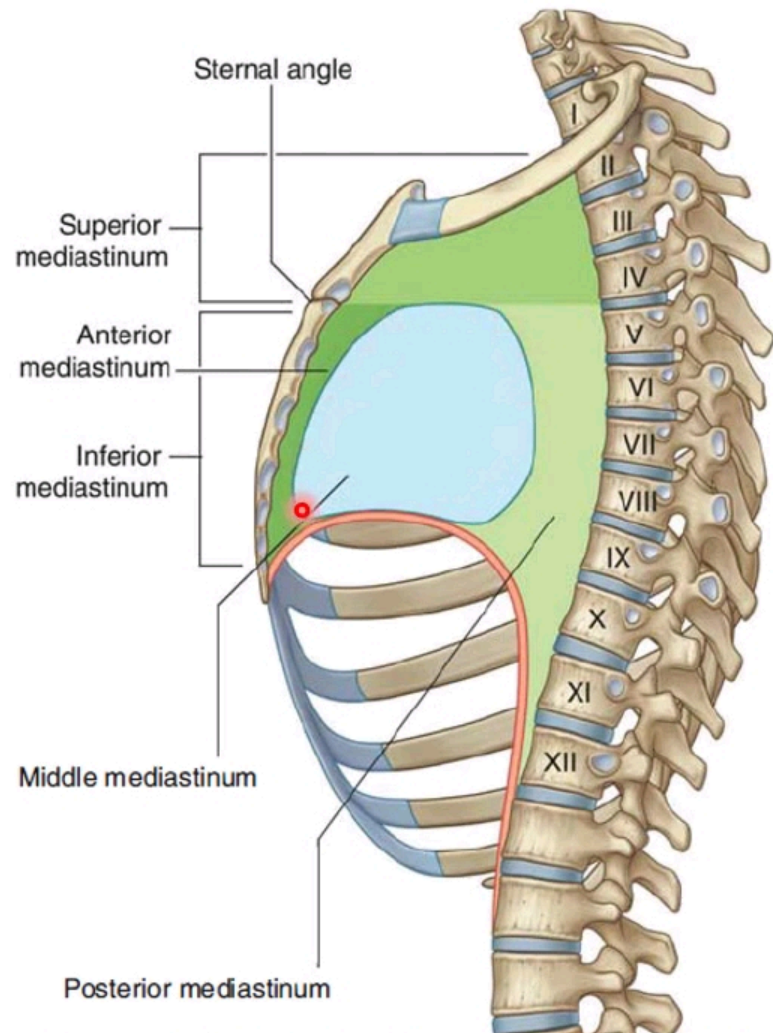
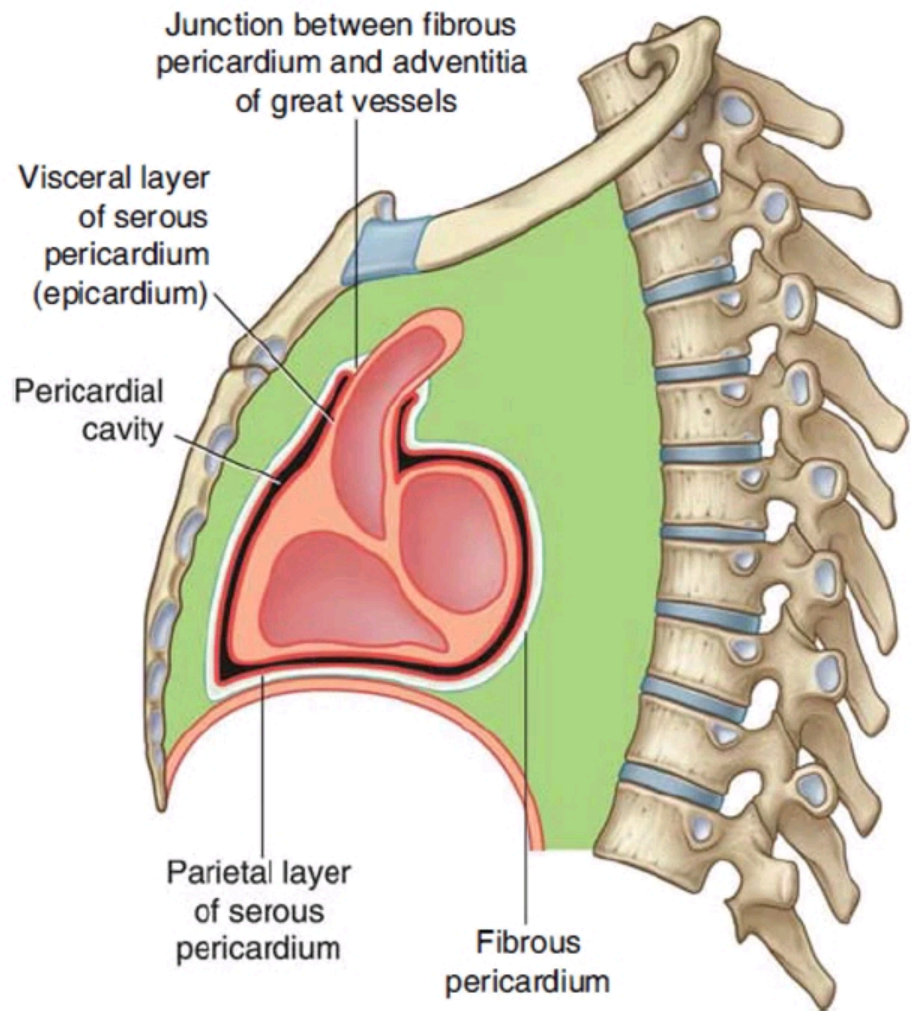
- Right atrium.
- Right Ventricles.
- Left atrium.
- left Ventricles.
- Aortic Valve .
- Pulmonary Valve .
- Cardiac Skeleton.
- Coronary Arteries.
 - Right Coronary Arteries.
 - Left Coronary Arteries.
 - Coronary Arteries Variation.
- Coronary Sinus.
- Conductive system of the heart.
- Cardiac Plexus
- Passage of Blood Through the Heart.
- location of the heart valves, and auscultation points.

Middle Mediastinum

The middle mediastinum (a term not frequently used) contains:

- ✓ Pericardium.
- ✓ Heart.
- ✓ The adjoining parts of the great vessels.
- ✓ The lung roots.
- ✓ The phrenic nerves.
- ✓ The deep part of the cardiac plexus.



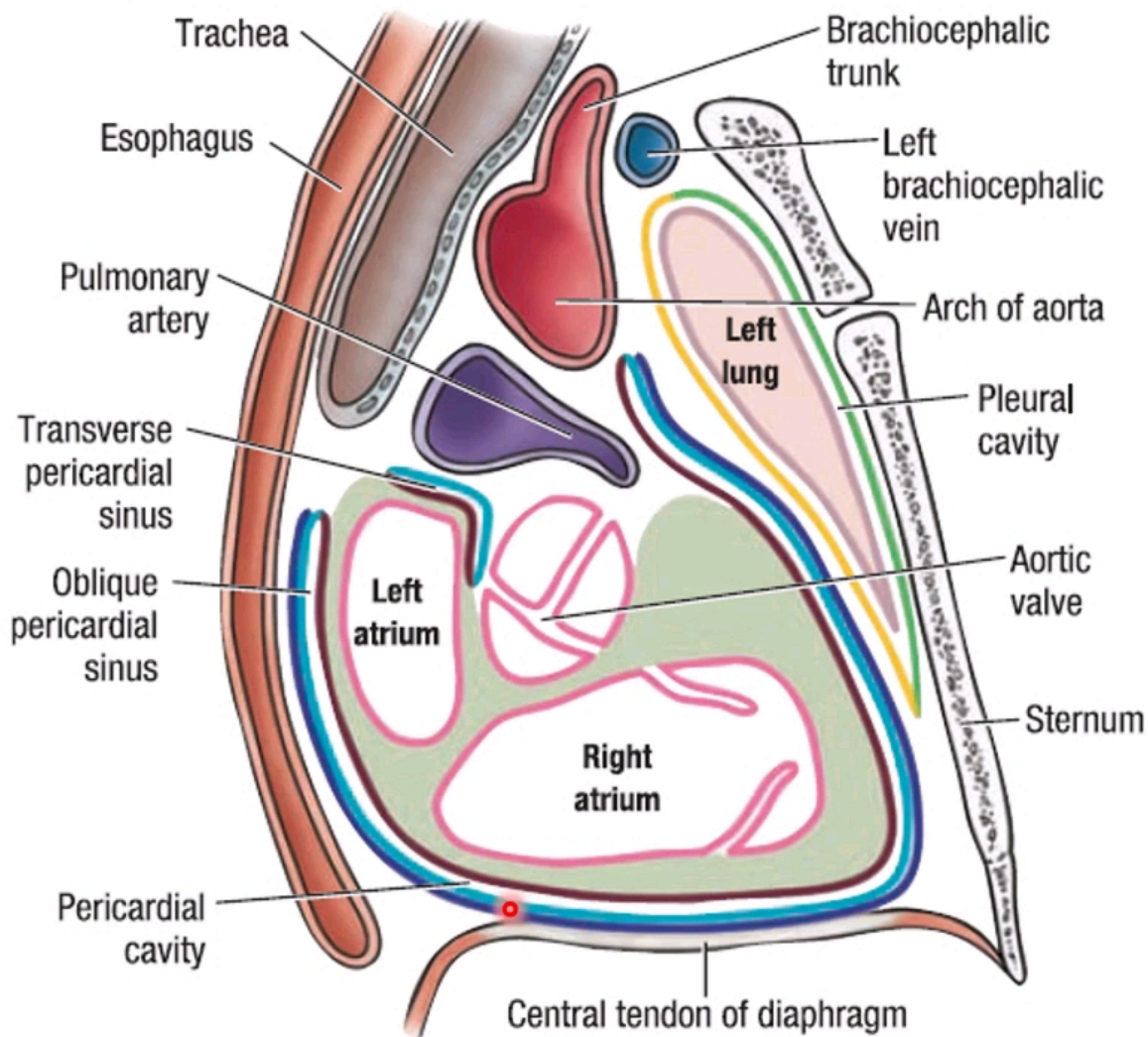


The Pericardium

- It is **fibro-serous sac** surrounding the heart
- It surrounds the great vessels of the heart
- **It consists of:**
 - 1\ outer fibrous layer.
 - 2\ inner serous layer.

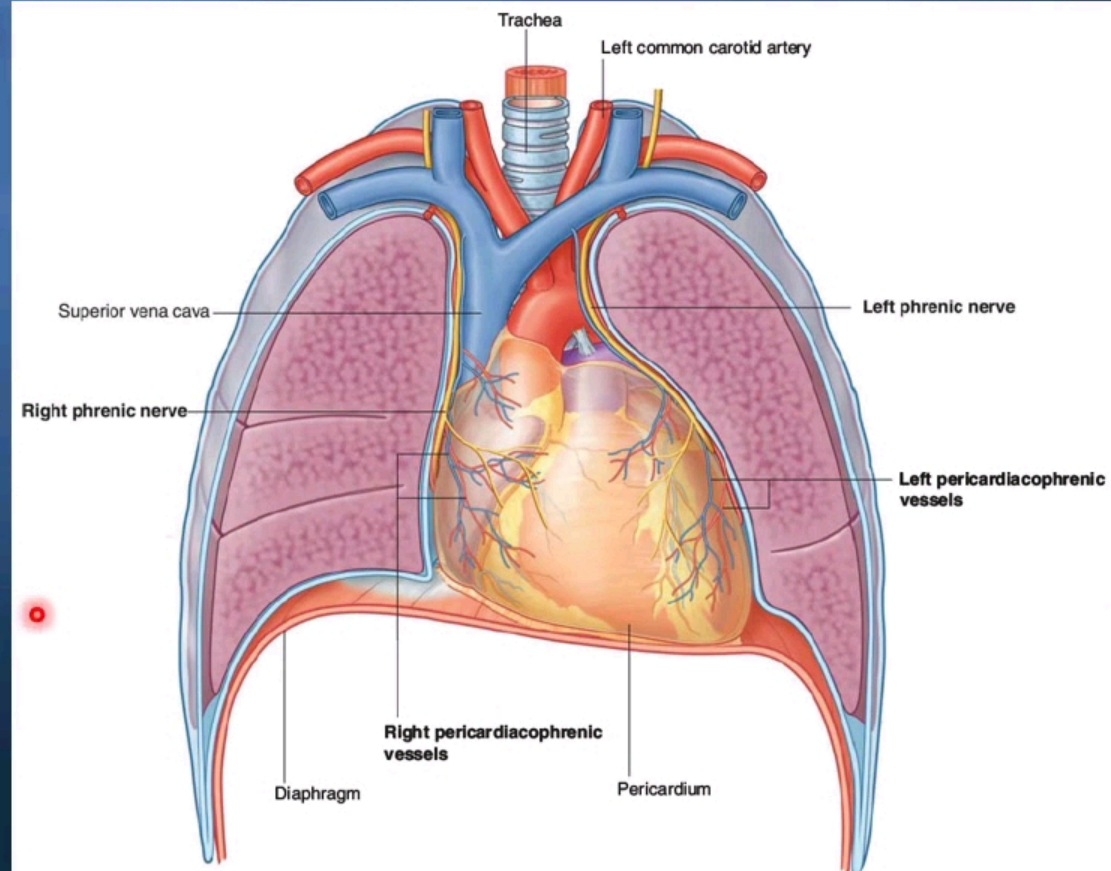
parietal

visceral



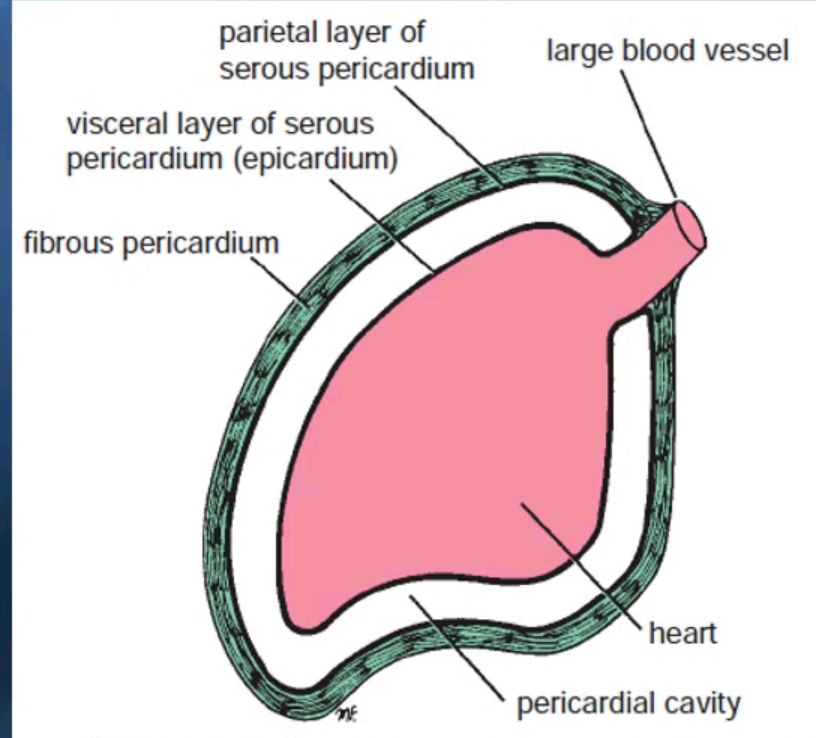
Pericardium

- It is a **fibroserous sac** that **encloses the heart and the roots of the great vessels** and occupies the **middle mediastinum**.
- It is composed of :
 - The fibrous pericardium.
 - The serous pericardium.
- It receives blood from:
 - The pericardiophrenic arteries.
 - The bronchial arteries
 - The esophageal arteries.
- It is innervated by:
 - The phrenic.
 - The vagus nerves.
 - the sympathetic trunks.



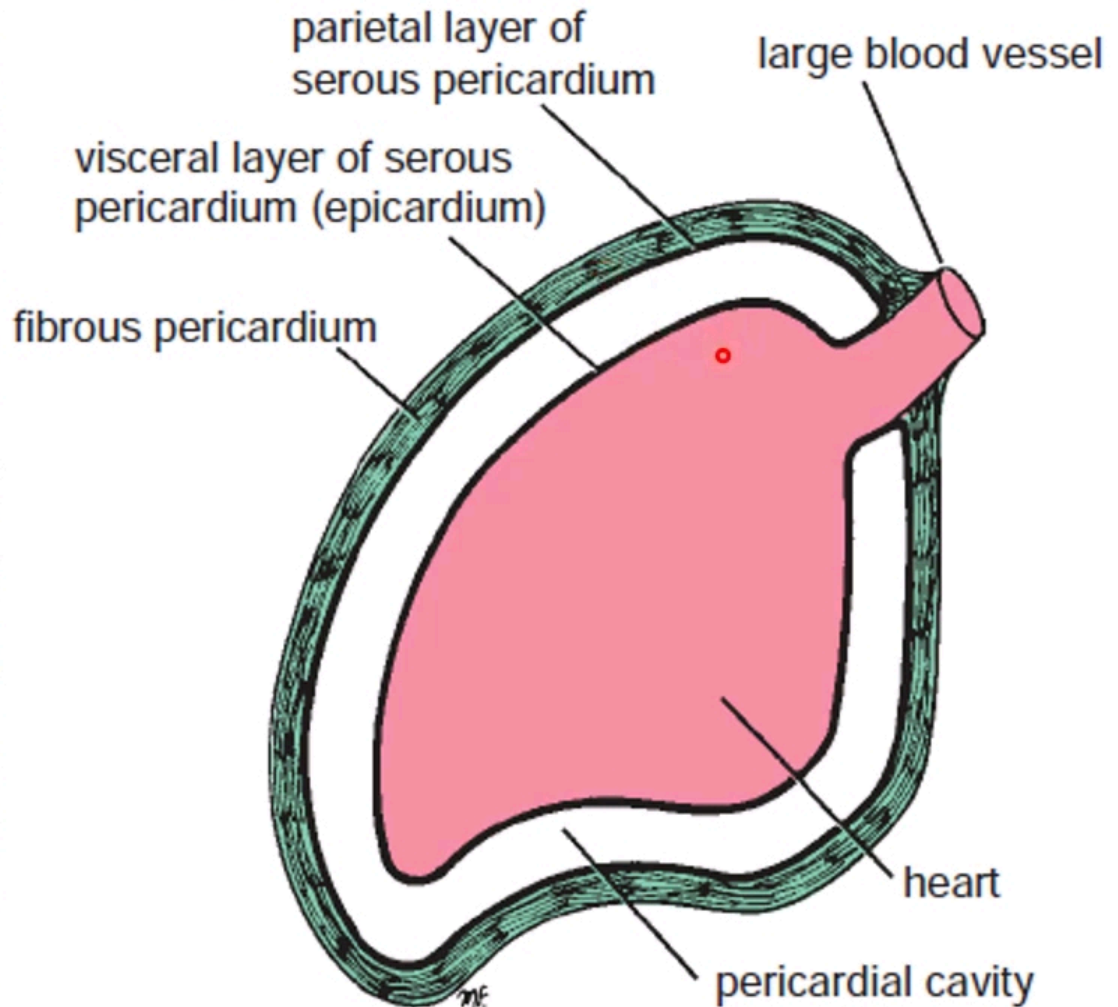
Fibrous pericardium

- It is **outer single-layered fibrous sac** that encloses:
 - The **heart**.
 - The **roots of the great vessels**.
 - It is fusing with the adventitia of these vessels.
- Its **broad base** overlies **the central tendon of the diaphragm**.
- The **phrenic nerves** lie on the surface of **the fibrous pericardium** and the mediastinal pleura is adherent to it.
- It is connected to the back of the sternum by weak **sternopericardial ligaments**.
- It is supplied with blood by the **internal thoracic arteries**.



Serous Pericardium

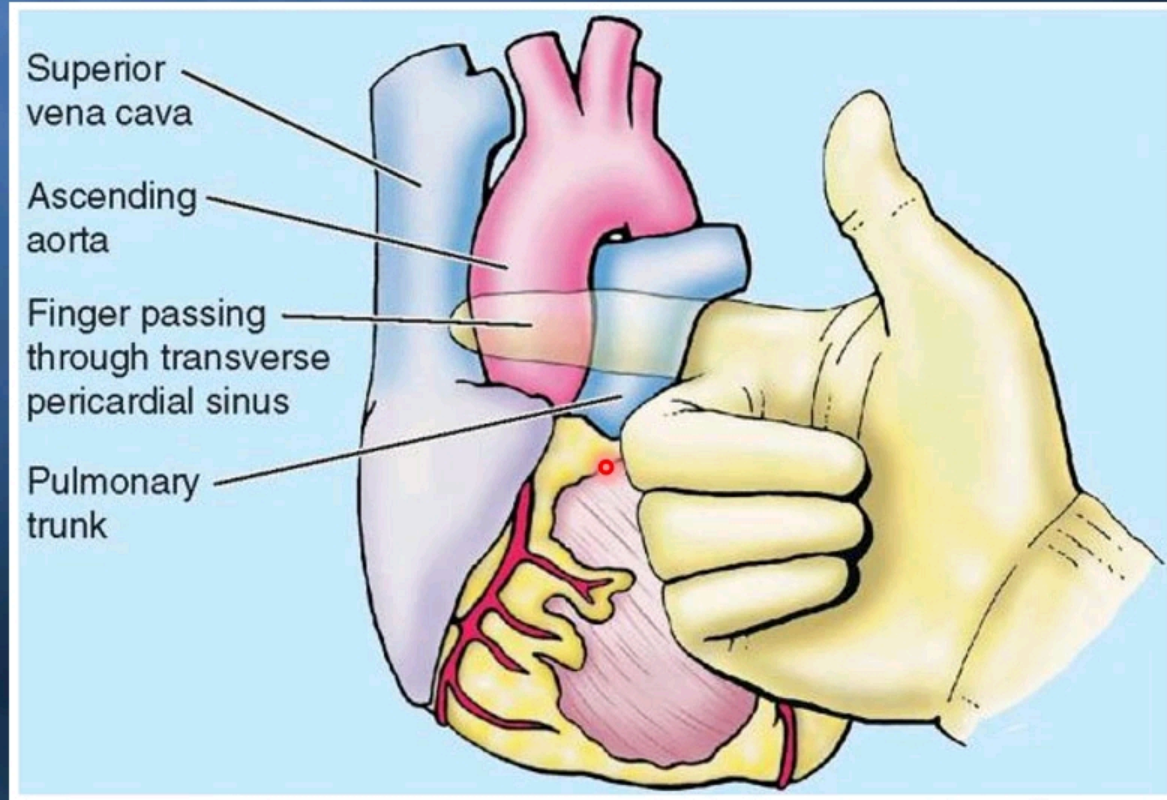
- It consists of:
 - **The parietal layer**
 - It lines the inner surface of the fibrous pericardium.
 - **The visceral layer**
 - It forms the outer layer (epicardium) of the heart wall and the roots of the great vessels.



Pericardial Sinuses

1) Transverse Sinus:

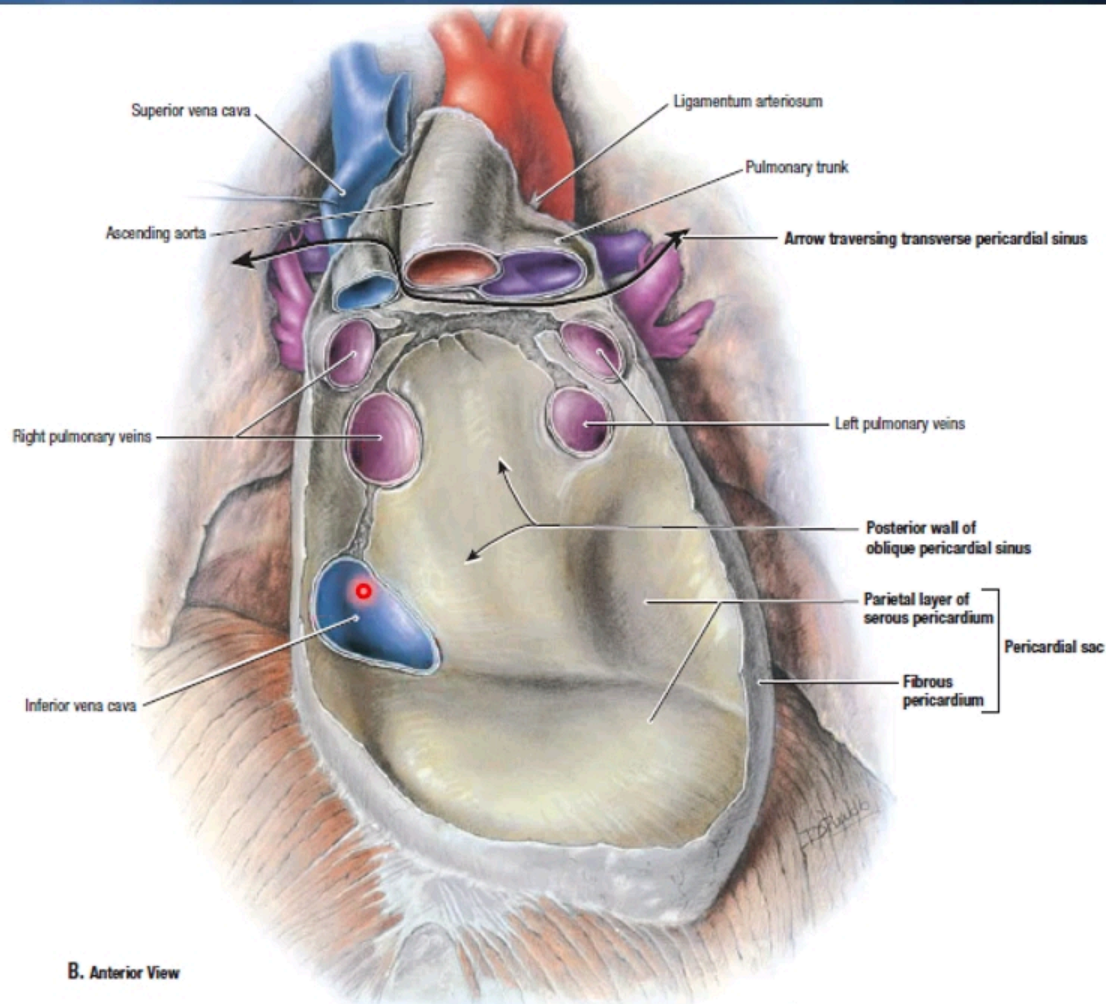
- It is a subdivision of the **pericardial sac**.
- It is lying posterior to the ascending aorta and pulmonary trunk, anterior to the SVC, and superior to the left atrium and the pulmonary veins.

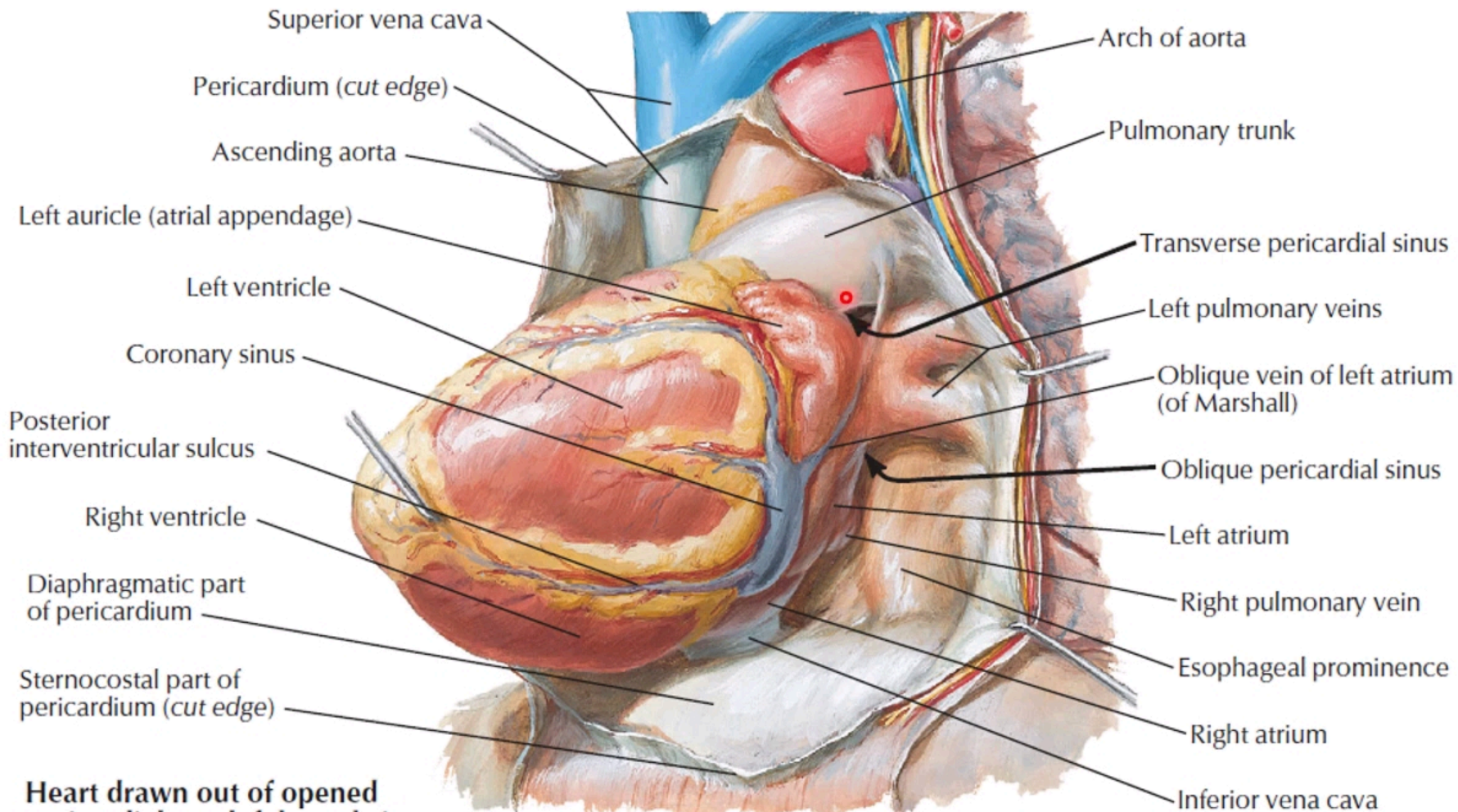


Pericardial Sinuses

2) Oblique Sinus:

- It is a subdivision of the **pericardial sac** behind the heart.
- It is surrounded by the reflection of the serous pericardium around the right and left pulmonary veins and the inferior vena cava (IVC).





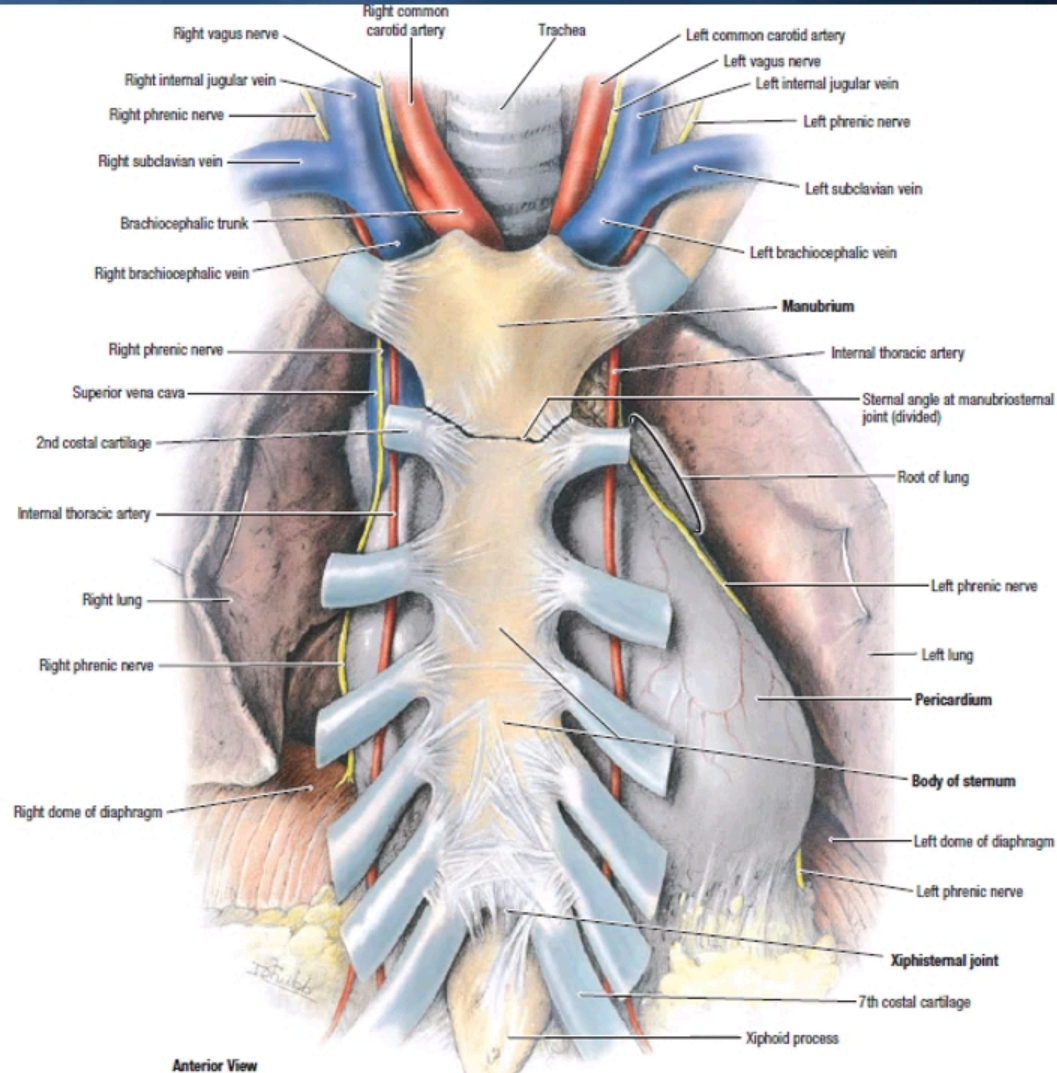
Heart drawn out of opened pericardial sac: left lateral view

Nerve supply

- The fibrous pericardium is supplied by **the phrenic nerve**.
- The parietal layer of serous pericardium is supplied by **the phrenic nerve**.
- The visceral layer of serous pericardium on the heart surface is insensitive.

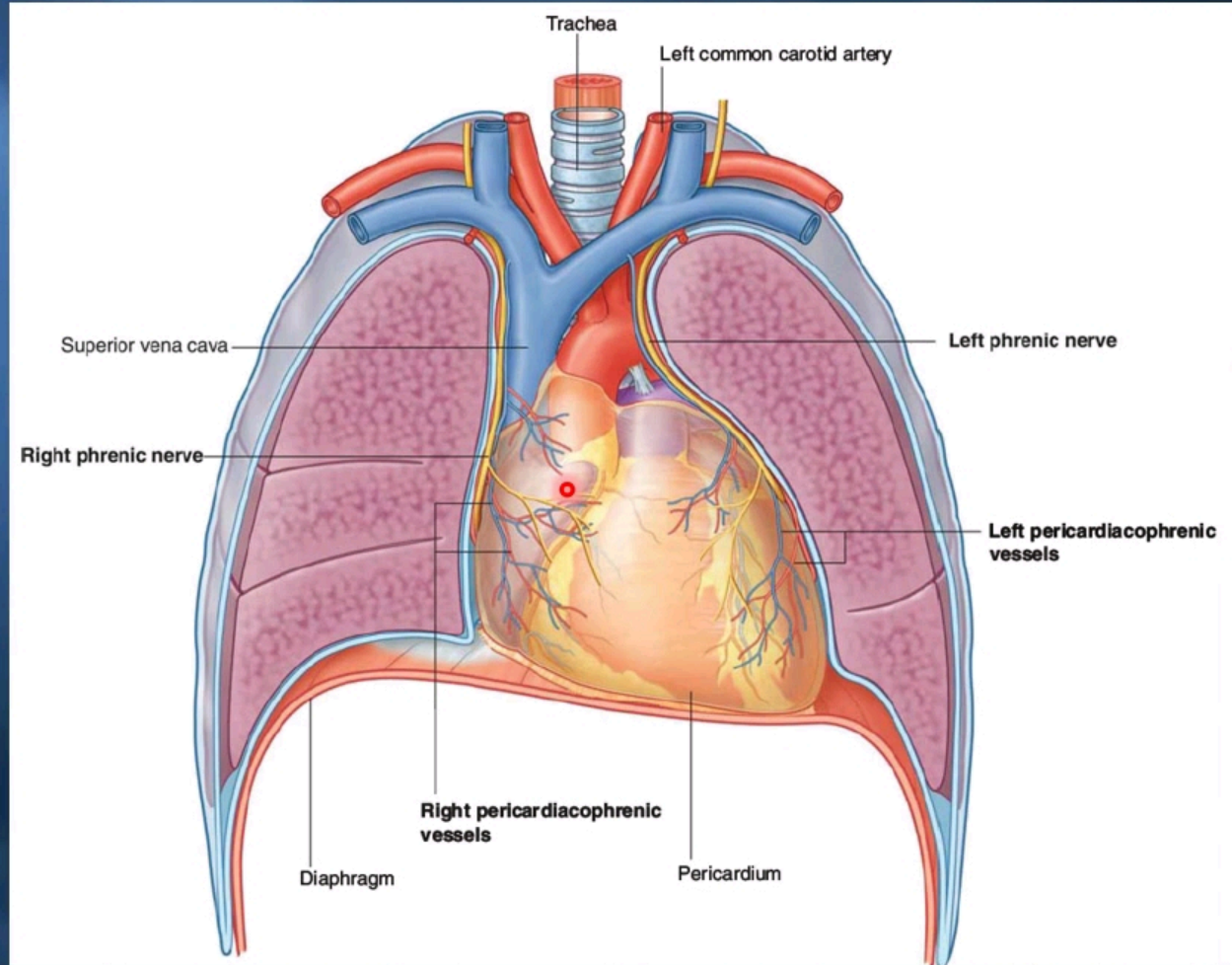
Note:

- Pain from the heart (angina) is transmitted by **sympathetic**
- The pain of pericarditis originates in the **parietal layer only**, and is transmitted by **the phrenic nerve**.

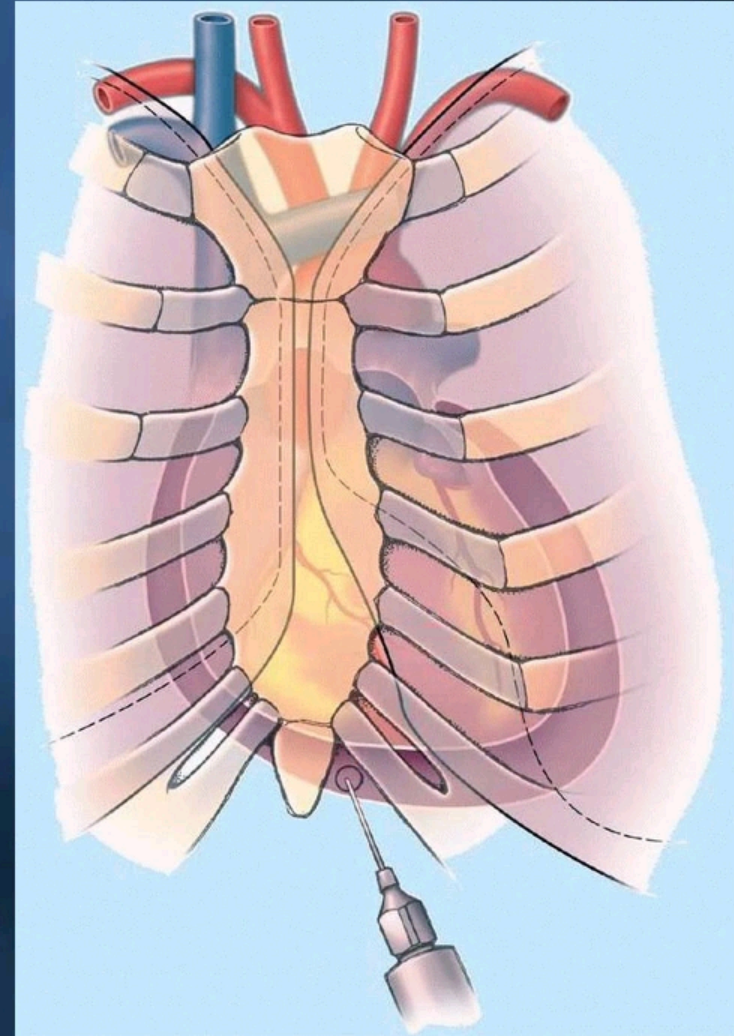
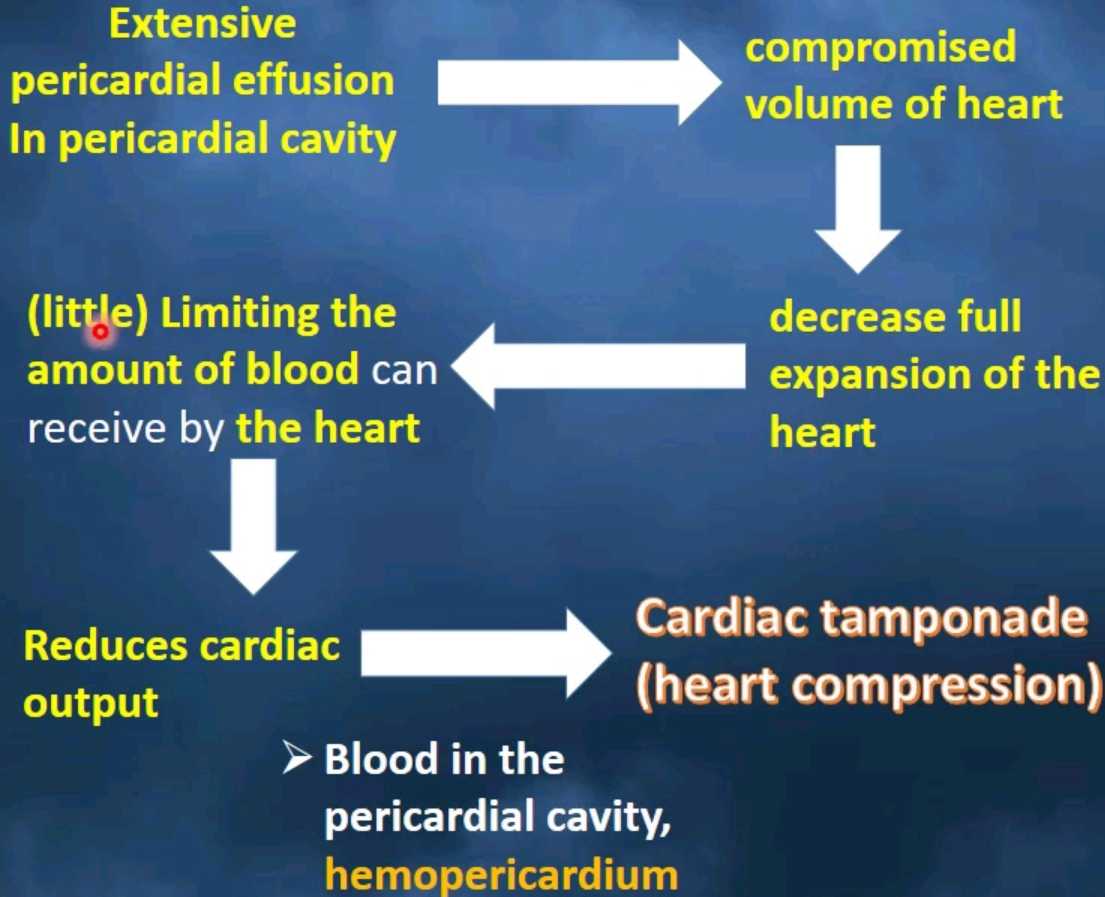


Pericardial Blood supply

- It is derived from **the internal thoracic artery**:
 - Pericardiophrenic artery.
 - Musculophrenic branches.
 - Bronchial arteries.
 - Thoracic aorta.
- The veins drain into **the azygos system**.



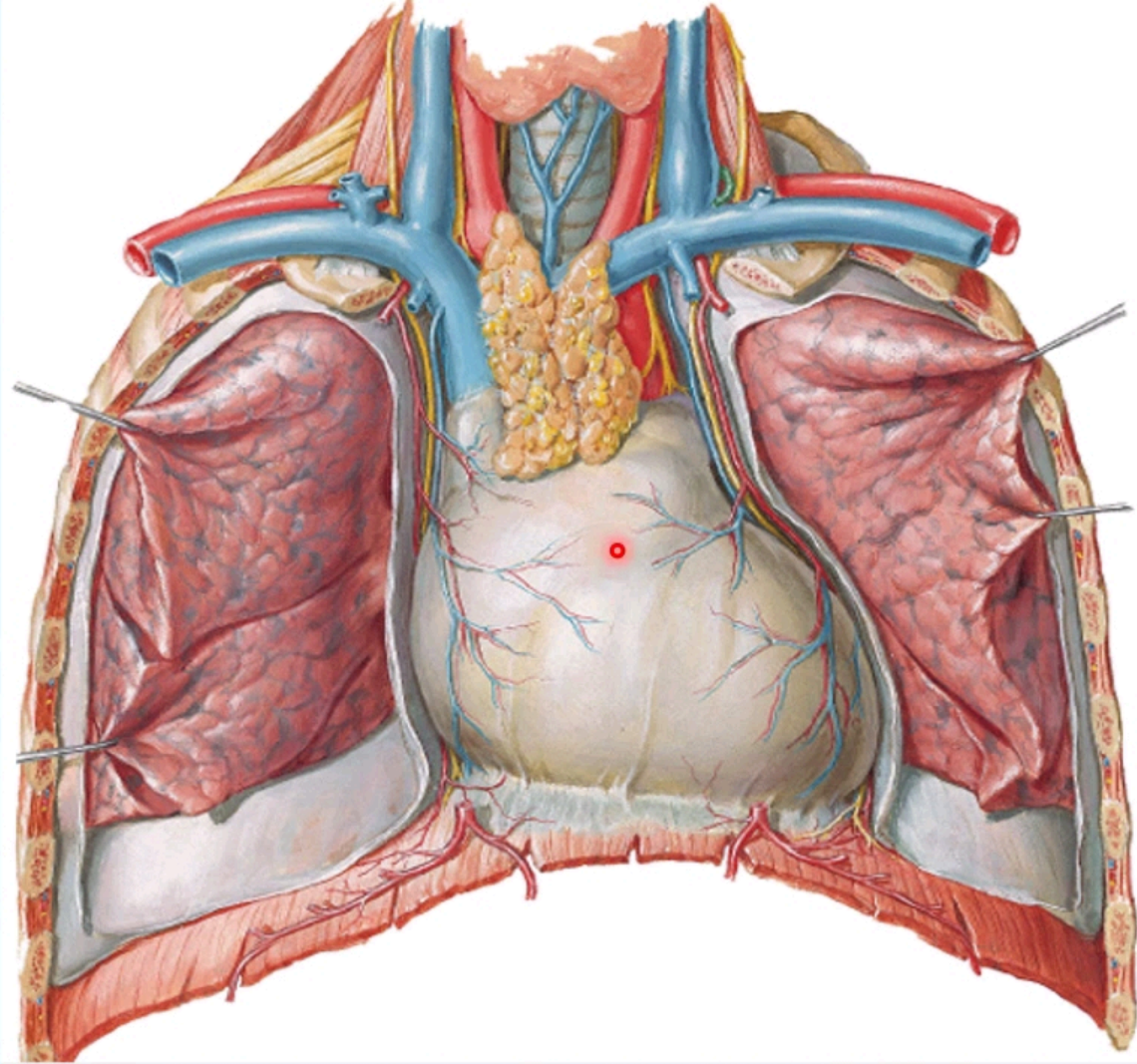
Pericardiocentesis



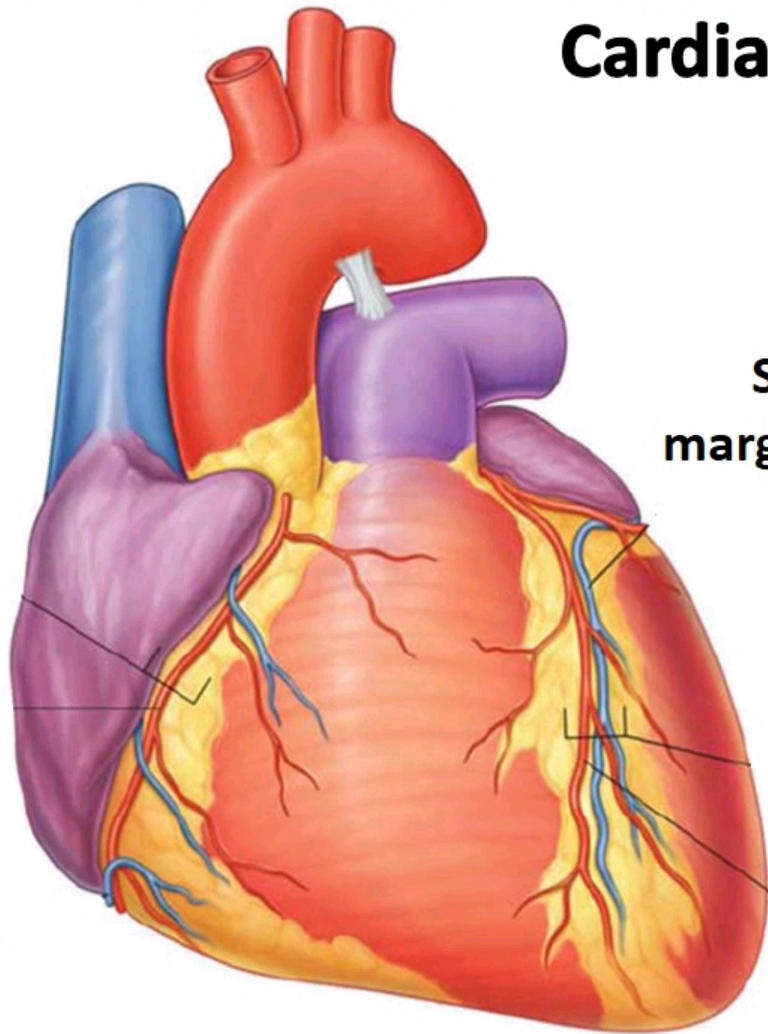
HEART

Location of the heart

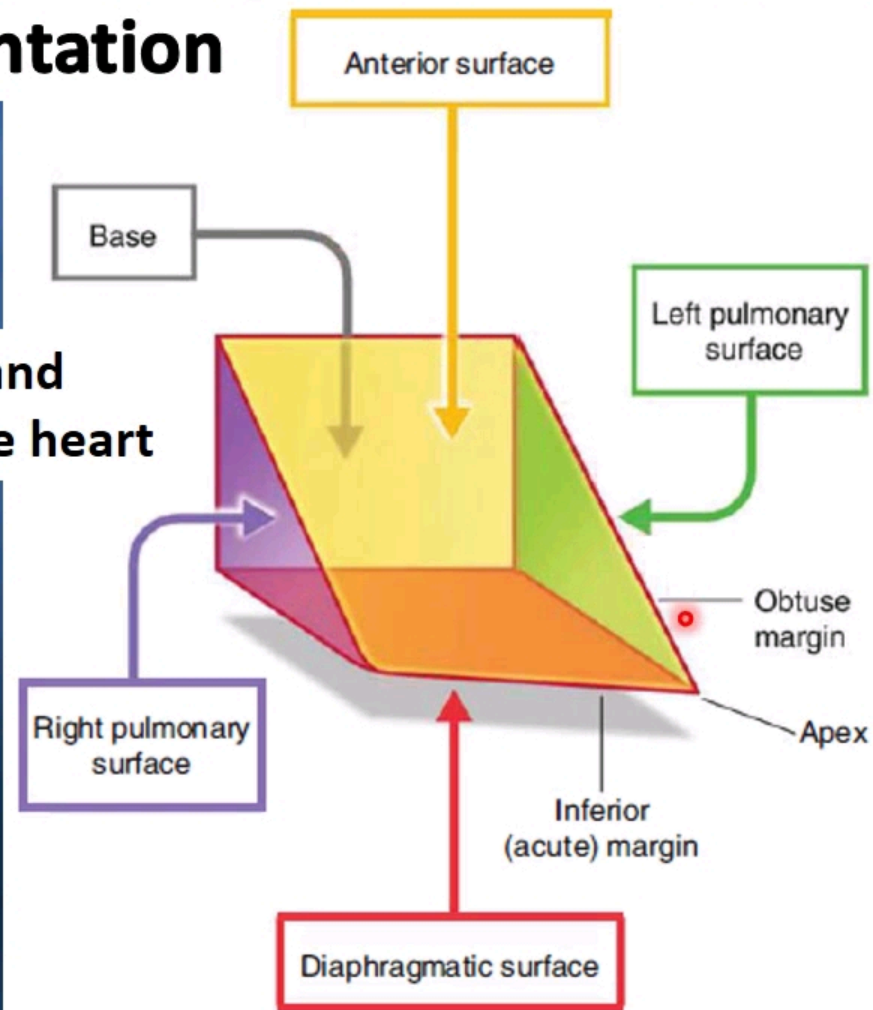
- ✓ The heart lies obliquely in the thorax.
- ✓ its long axis passing downwards and to the left to the apex.
- ✓ The heart has undergone a degree of rotation during development,
- ✓ the chambers are not located strictly in the positions that their names suggest.
- ✓ The right-sided chambers are mainly anterior to their left-sided counterparts.
- ✓ the heart is described as having:
 - ✓ Right border.
 - ✓ inferior (acute) border.
 - ✓ left (obtuse) border.
 - ✓ Anterior (sternocostal) surface,
 - ✓ Inferior (diaphragmatic) surface,
 - ✓ Base or (posterior) surface,
 - ✓ Apex at its left inferior corner.



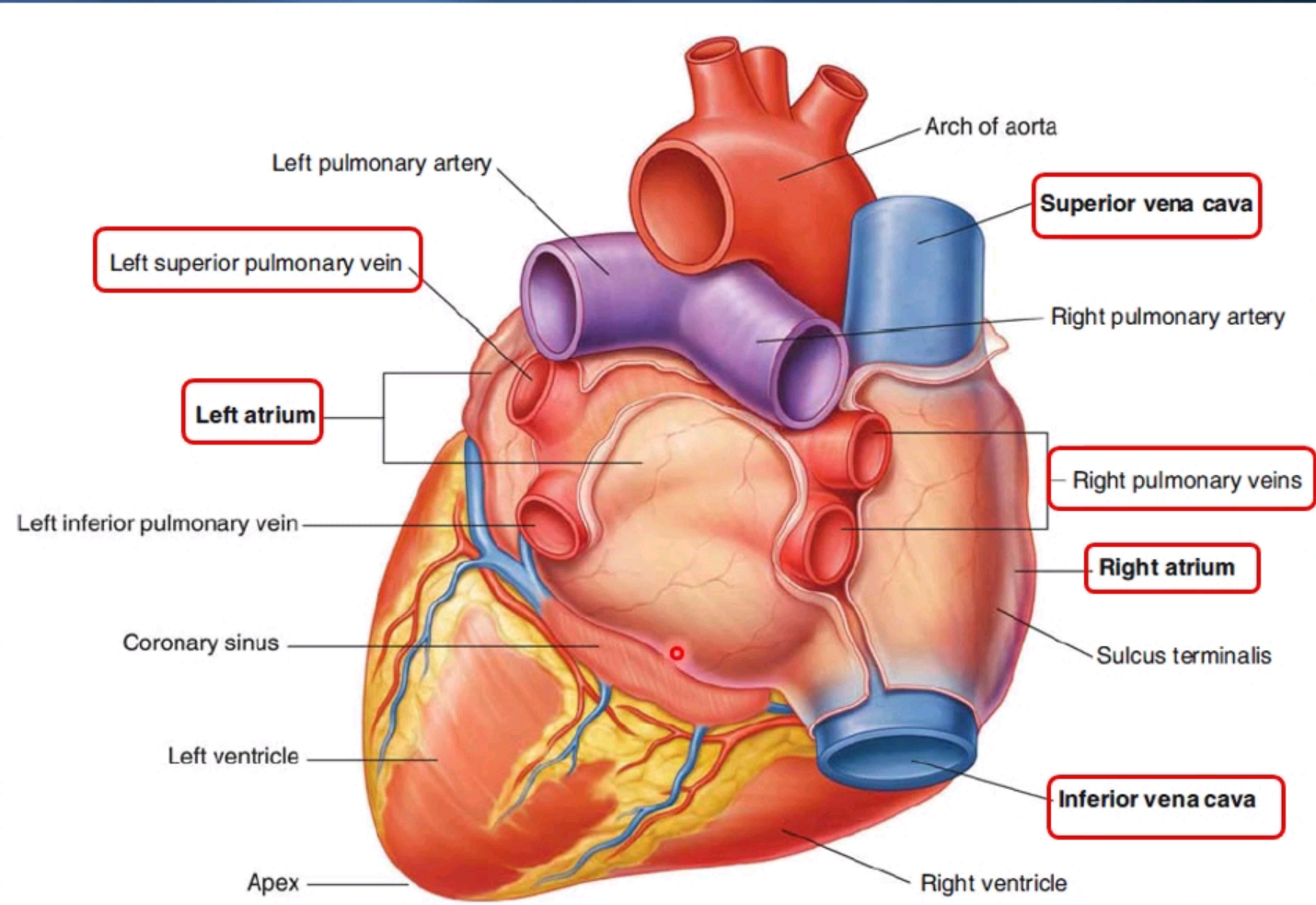
Cardiac Orientation



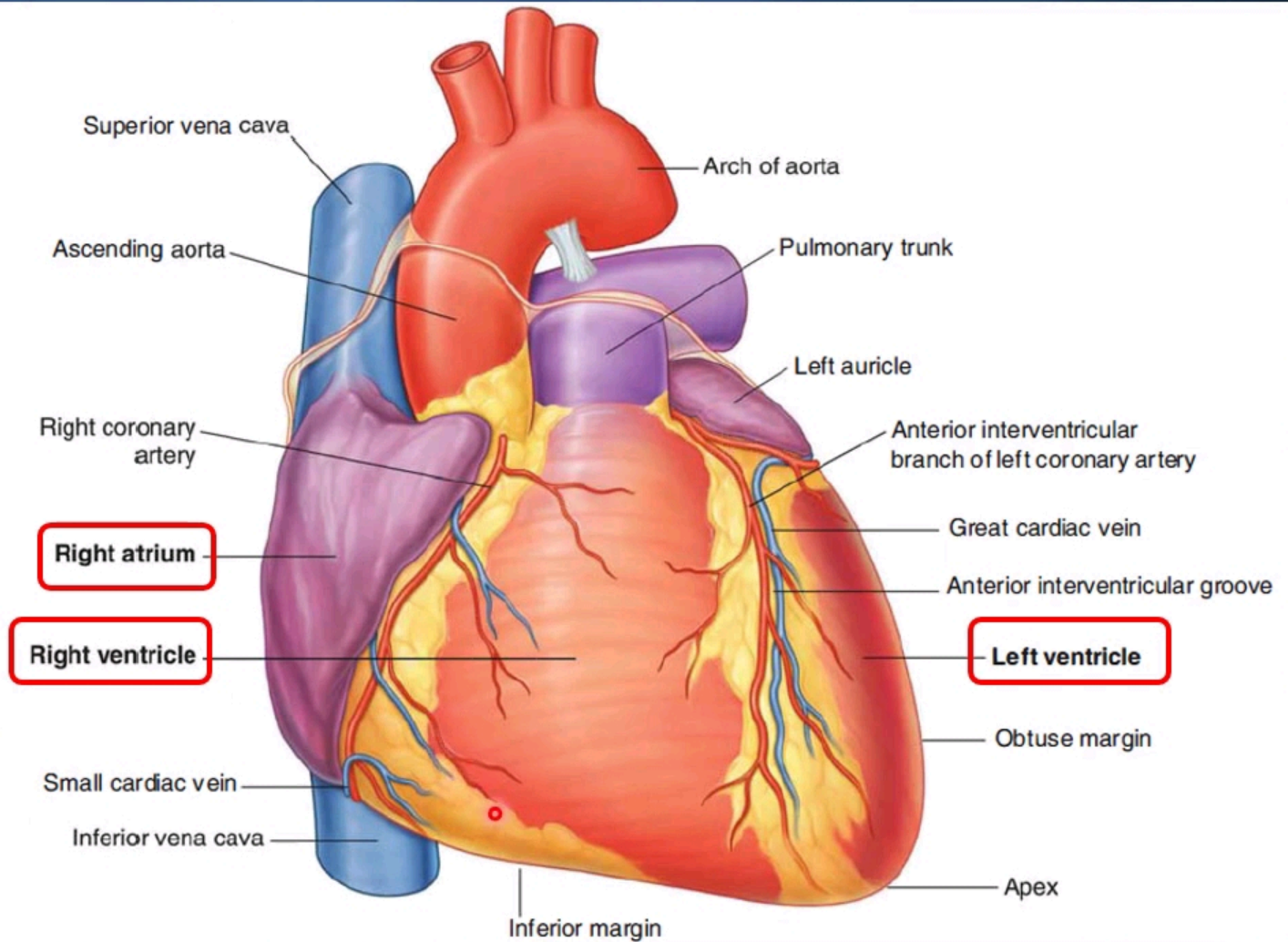
Surfaces and margins of the heart



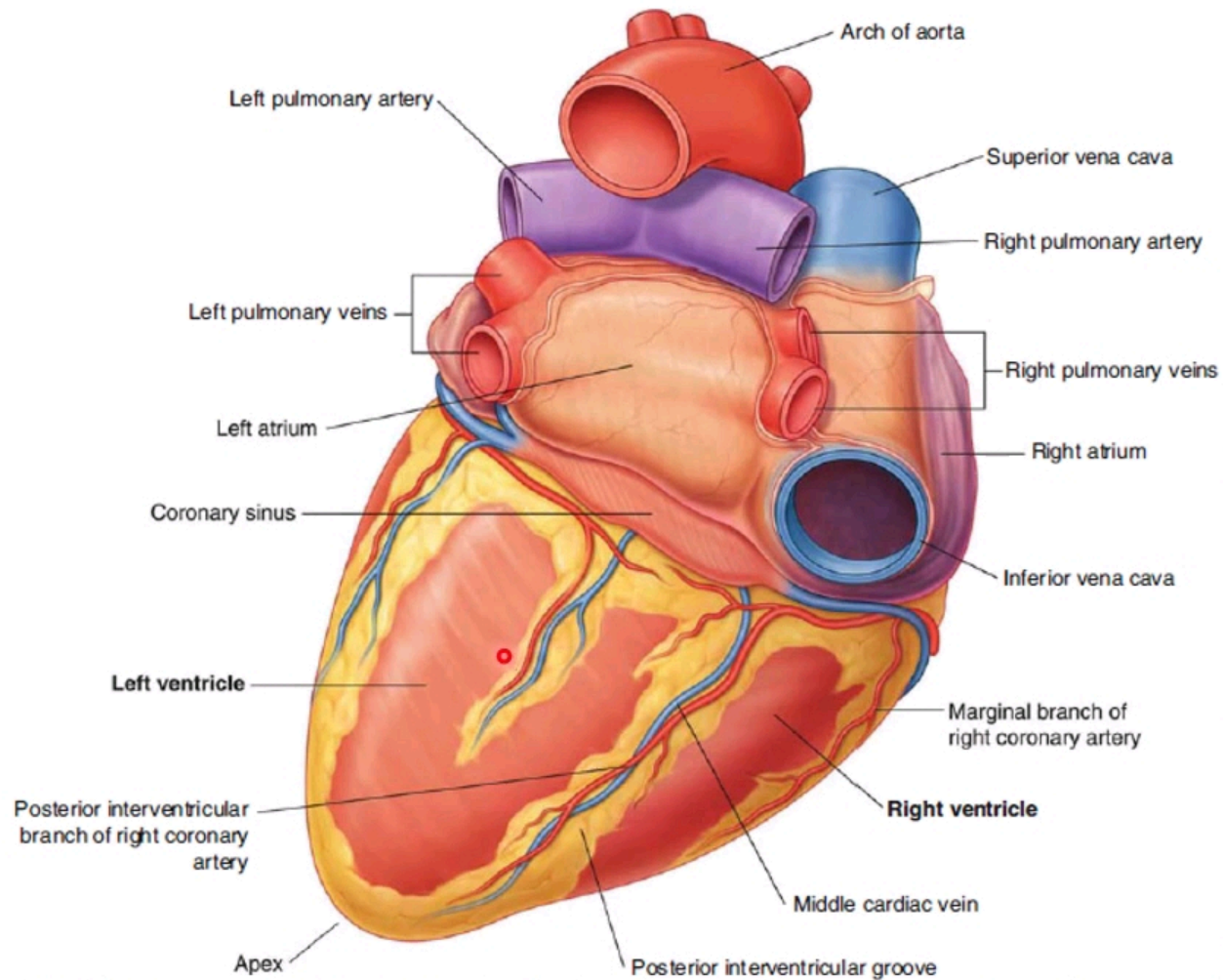
Base of the heart



Anterior surface of the heart



Diaphragmatic surface

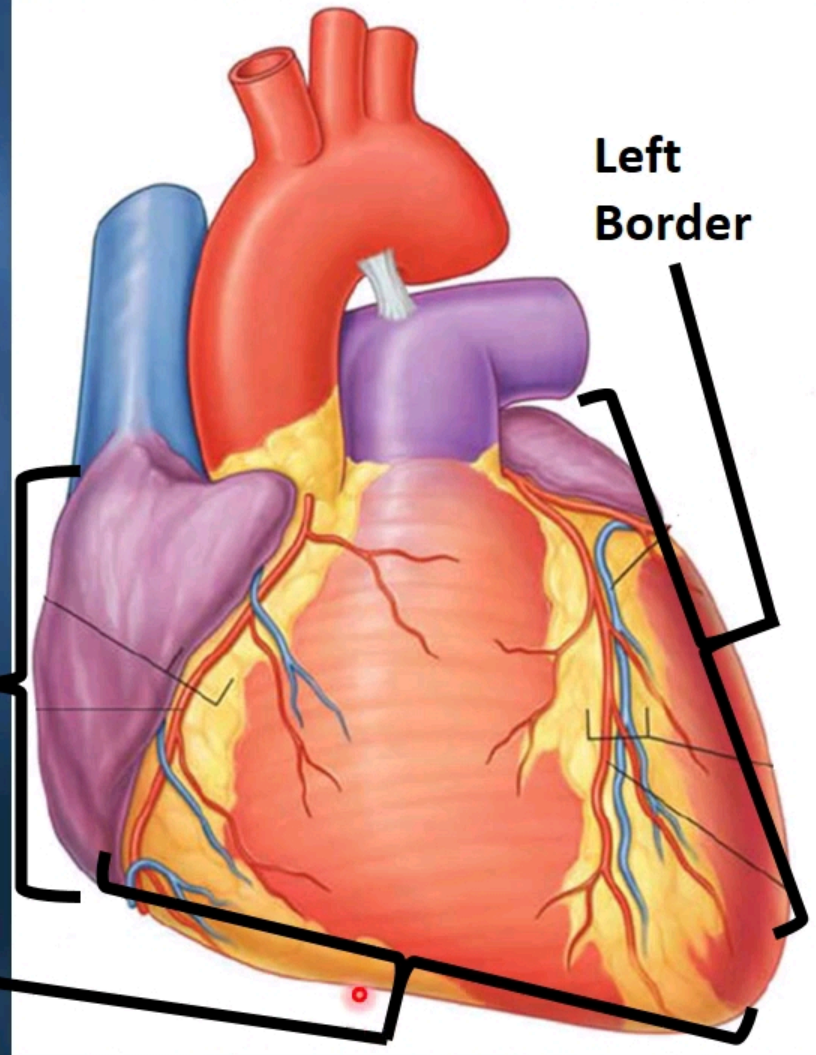


The Border and margin of the heart

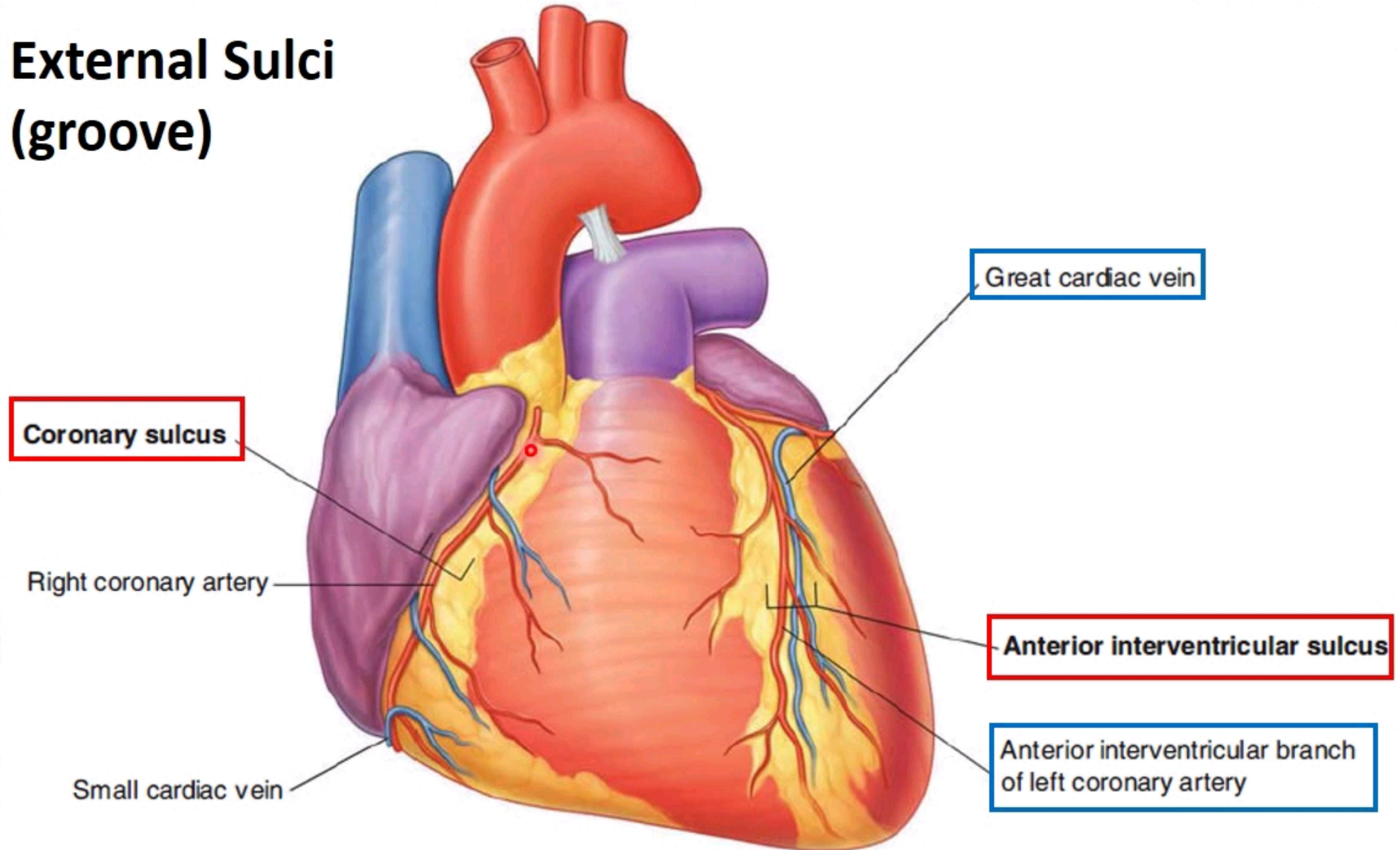
Right Border

Inferior Border

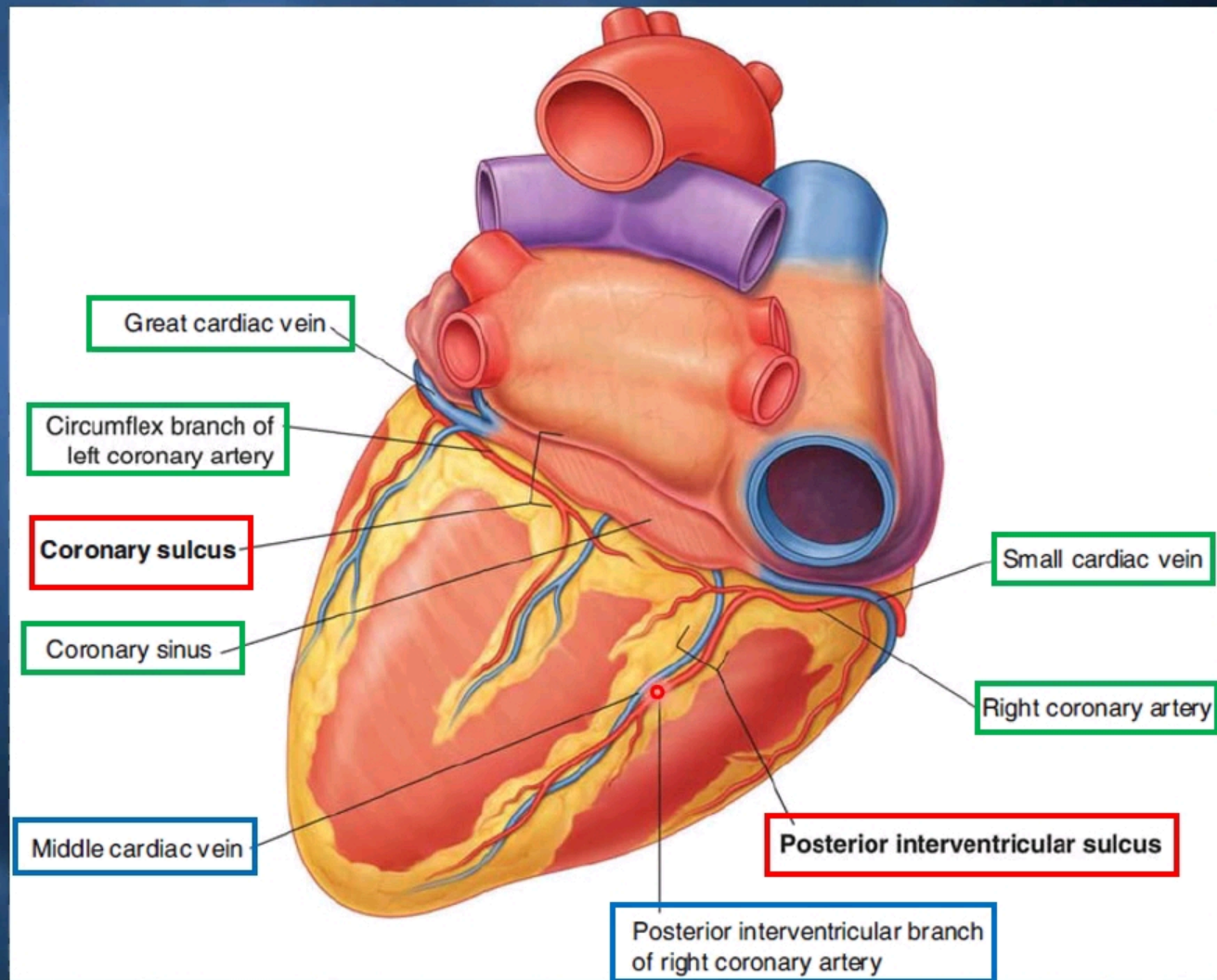
Left Border



External Sulci (groove)



External Sulci (groove)





Thanks